The effect of baffles in inhibiting heat transfer through an interface under different sloshing conditions

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Abstract

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1. Introduction

Having been conceived and constructed a century ago [1, 2], the liquid propellant rocket engine has played a pivotal role in the annals of human spaceflight history due to its high-thrust performance and high combustion energy-to-mass ratio. Given the intricate nature of liquid rocket engine design and the demanding operational conditions, engineers from various nations and regions have diligently worked towards its optimization and enhancement [1, 2, 3, 4, 5]. Nevertheless, the history of human aviation has witnessed numerous launch mission failures, highlighting the ongoing need for improvement. Notably, propellant and oxidizer together constitute over 80% of a liquid rocket engine's total mass. The varying states of these substances during storage, liftoff and flight play a crucial role in determining the success of a rocket mission. The storage of a significant quantity of liquid propellant in cryogenic tanks presents notable technical hurdles, including issues like thermal stratification [6, 7], pressure drops [8, 9], and even the possibility of ullage gas collapse [10, 11] within the cryogenic containers. These phenomena are all related to the heat and mass transfer in the cryogenic tank, especially when the tank experiences severe sloshing, amplifying the impact of the above phenomena on the stability of the cryogenic liquid propellant systems of the rocket. In addition, the substantial volume of liquid introduces a heightened susceptibility to any form of motion, which

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can significantly influence the aircraft's trajectory. Therefore, during the early stages of liquid rocket engine development, researchers had already undertaken experimental[12] and numerical[13] studies on sloshing behavior within spherical tanks.

Hence, numerous scholars have conducted research on sloshing behavior under various conditions. Kana [14] conducted an analytical study on a nonlinear rotary sloshing in propellant tanks. Eswaran [15] did a comprehensive review of several experimental studies addressing sloshing issues in partially filled tanks. He pointed out that experimental research could help to check the validation of assumptions of mathematical models, numerical simulations, and to employ the model effectively for design applications. Nonetheless, owing to the intricate nature of the experiment, constraints associated with detection methods, significant expenses and extended experimental cycles, an increasing number of scholars have shifted their focus on studying sloshing behavior through numerical methods. The sloshing motion is a profoundly nonlinear and stochastic process and brings a significant challenge to traditional CFD tools. Agarwal[16] employed the Volume-of-Fluid (VOF) method within ANSYS FLUENT to capture the free surface and predict the formation of gas-core vortices during the discharge of liquid propellant. By integrating the VOF method with the moving mesh technique, Liu [17] attained a heightened level of precision in capturing the free surface motion while investigating the hydrodynamic behavior of sloshing within a hydrogen tank. Hari [18] incorporated the Arbitrary Lagrangian-Eulerian (ALE) approach into his finite element method (FEM) to model the deformation of the elastic tank and subsequently assessed the influence of the elastic wall on the tank's sloshing behavior. Due to the inherent characteristics of precisely capturing the material interfaces of Smooth Particle Hydrodynamics (SPH), more scholars choose to use the SPH method to study complex tank sloshing behavior. Cao [19] employed the SPH method in the research on the characteristics of the sloshing loads in a tank. Green [20] use a δ-SPH method for a long-duration violent three-dimensional sloshing research. Through the integration of the Particle Shifting Technique (PST) and an enhanced multi-phase SPH model with a pressure correction algorithm, Yang [21] successfully replicated the occurrence of the gas phase becoming entrapped within the internal water flow and undergoing movements such as rolling and rising during a two-phase sloshing scenario. Simultaneously, numerous scholarly works have started addressing strategies for mitigating tank sloshing to minimize its influence on liquid rocket engines. Cole [22] measured the damping forces on fuel sloshing baffles of varying thickness in a very early stage. Himeno [23] combined experimental and numerical methods to verify that the baffle in a reusable launch vehicle can effectively resist the sloshing of liquid and is expected to prevent catastrophic gas inhalation, even in harsh environments where vertical acceleration is comparable to lateral acceleration. Pan [24] introduced a floating baffle and highlighted that this movable structure can elevate the frequency of liquid sloshing, efficiently dissipate sloshing kinetic energy and diminish the amplitude of liquid sloshing. Konopka [25] introduced a six-legged anti-sloshing system capable of implementing active damping. Combined with its control algorithm,

it can suppress linear lateral sloshing in two cycles within 4 seconds, which is planned to be integrated into future upper stages.

Besides investigating the sloshing behavior itself, comprehending the heat and mass transformations that take place during the sloshing process is crucial in understanding various potentially hazardous phenomena in cryogenic tanks. One prominent area of heat transfer focuses on the thermal stratification [26, 27, 28] effect within cryogenic tanks, which arises from natural convection. Besides, there are scholars who worked on the isolation or non-isolation effect of heat transformations. Joseph [28, 29] works on the effect of insulation thickness on pressure evolution, while Liu studies the phenomena of heat leakage [30] and heat invasion [31] respectively. Considering the phase transitions resulting from heat exchange processes, alterations in pressure within the cryogenic tank emerge as crucial elements influencing the performance of liquid rockets. Behruzi [32] proposed an improved phase change model to predict the lower pressure drop under a slight sloshing excitation using a grid-based numerical scheme. Their results matched the LN2 experimental carried out by ESA. Himeno [9] and Kartuzova [33] utilized the CFD experiments to investigate the phase change and the pressure drop caused by the heat transfer occurrences in a sub-scale cryogenic storage tank subjected to violent sloshing conditions. Montsarrat [8] pointed out that the sloshing-caused thermodynamic equilibrium is the most possible reason for the pressure drop of the LH2 tank in the upper stage of Ariane 5 in some ascent phases. Furthermore, Anderson [11] integrated a numerical tool with an analytical model to forecast evaporation and the ensuing ullage collapse within the propellant tank of an upper stage, which is subject to simulated orbital maneuvers under microgravity conditions. In studying the phenomena associated with heat and mass transfer mentioned above, scholars [9, 10, 32] have emphasized the close correlation between the observed occurrences and the heat exchange taking place across material interfaces within the cryogenic tank. Therefore, a quantitative study of heat transfer across the gas-liquid interface during sloshing in the cryogenic tank will provide crucial insights for guiding the design and performance-improving of such tanks. Nevertheless, obtaining quantitative data on heat transfer through the two-phase interface via experimental measurements proves challenging, and the traditional CFD method faces limitations in effectively tracking the dynamic wave surface. Considering the inherent capacity to effectively capture dynamic material interfaces of the Lagrangian-based algorithm, utilizing the SPH method appears to be a favorable decision for calculating heat transfer in the aforementioned scenario.

In this paper, we present a validated open-source SPH code (SPHinXsys) [34, 35] suitable for simulating multiphase sloshing in cryogenic storage tanks, along with modeling the heat exchange phenomena during the sloshing process. Meanwhile, we have investigated heat transfer across the material interface under various sloshing conditions, including different frequencies, amplitudes, and the presence or absence of baffles, and have derived conclusions regarding the relationship between sloshing dynamics and heat transfer mechanisms. Initially, we present a detailed overview of the numerical algorithms employed in this study and propose an effective approach to mitigate numerical instabil-

ities encountered during high-resolution simulations of two-phase flows using particle-based methods. Subsequently, through comparisons with experimental data and other numerical methods, we validated the ability of our code to accurately track the evolution of the free surface during intense sloshing events, as well as its precision in calculating two-phase heat transfer. Finally, we calculate the heat transfer across the material interface in a realistic rocket tank containing liquid methane and nitrogen under various sloshing modes and provide a qualitative analysis of the results. The results indicated that when the sloshing frequency closely aligns with the natural frequency of the tank, there is no notable decrease in the heat flux crossing the material interface as the sloshing process progresses in specific periods. In contrast, the sloshing under the frequency away from the natural frequency had a significant heat flux decay as sloshing continued. In addition, we assessed the influence of various baffle configurations on heat transfer. The calculations revealed that a perforated baffle could efficiently dampen sloshing and impede heat transfer across the two-phase interface during sloshing. Nevertheless, the introduction of baffles alters the natural frequency of the tank, resulting in corresponding variations in the heat flux patterns through the interface.

The remainder of this paper is organized as follows. In Section 2, we briefly summarize the governing equations and the methodology in SPHinXsys we employed for two-phase sloshing simulation in this paper, and An improved bounding method is introduced to prevent the dissolution of free gaseous particles into liquid phases and their subsequent penetration through walls in contact with the liquid during high-resolution simulations. In Section 3, a 3D sloshing validation case with a two-phase heat transfer case are illustrated to prove that our code can not only preciously capture the dynamic evolution of the interface in a severe sloshing condition but also accurately calculate the heat transfer between the two different materials. A detailed calculation of heat flux through the interface between the liquid methane and gas nitrogen in a real rocket under different sloshing conditions is provided in Section 4. In Section 5, two distinct types of baffles were employed to assess their efficacy in attenuating sloshing behavior and impeding interfacial heat transfer. Conclusion remarks are given in Section 6. The code of the present method is available in our open-source SPHinXsys library [34, 35] at `https://www.sphinxsys.org`.

2. Methodology

In this section, we summarize the methodologies employed in this paper, including the governing equations and special numerical algorithms to facilitate the calculation of two-phase flow of tank sloshing.

2.1. Governing Equations

Both liquid and gas exist in real tanks, and in order to match the experiments used for verification, in this paper, we employed a viscous and incompatible two-phase flow. The density ratio of fluid to gas, i.e., $\rho_L/\rho_G >> 1$ with

the subscripts L and G represent liquid and gas respectively. The mass and momentum conservation equations can be written as

$$\begin{cases} \frac{\mathrm{d}\rho}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v} \\ \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{1}{\rho}(-\nabla p + \eta\nabla^2\mathbf{v}) + \mathbf{g} + \mathbf{f}^{s:p} + \mathbf{f}^{s:\nu} \end{cases}, \tag{1}$$

where $\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla$ stands for the material derivative, while ρ the density, $\mathbf{v}$ the velocity, p the pressure, $\mathbf{g}$ the gravity, η the dynamic viscosity, $\mathbf{f}^{s:p}$ the pressure force acting on fluid from the rigid wall and $\mathbf{f}^{s:\nu}$ the viscous force acting on the fluid from the rigid wall. Following the Ref. [36, 37, 38], the compressible effect of the light phase could be neglected, thus, an artificial isothermal equation of state (EoS)

$$p = {c^0}^2(\rho - \rho^0), \tag{2}$$

is adopted both for the heavy and light phases to close the equation set Eq. 1 with the weakly-compressible assumption [39, 40]. Here the ρ^0 is the initial reference density, while the $c^0 = 10 \times U_{max}$ is the artificial speed of sound relating to the maximum anticipated flow speed U_{max}.

Since the aim is to investigate heat flux across the two-phase interface, the heat transfer equation must also be taken into account. As the convective heat transfer component is inherently accounted for in the governing equations expressed in Lagrangian form when considering particle temperature, it is sufficient to focus solely on the thermal diffusion effect when calculating temperature variations. The equation of thermal diffusion can be written as:

$$\rho\frac{\mathrm{d}U}{\mathrm{d}t} = \nabla\cdot(k\nabla T), \tag{3}$$

where U is the internal energy, k the thermal conductivity coefficient. It can also be expressed in terms of temperature:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \nabla\cdot(\alpha\nabla T), \tag{4}$$

where T is the temperature of the particle, $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity. Here the c_p and k are the specific heat capacity and thermal conductivity, respectively. This solution has been successfully used in many particle-based simulations in industrial applications [41, 42, 43]. It has also been pointed out that this solution is particularly suitable for industrial applications where gravity or imposed velocity dominates[43].

2.2. Physics-driven particle relaxation

An important prerequisite for particle-based simulation is to obtain isotropic and body-fitted particle distribution for complex geometries, refers to the tanks and buffers in this paper. We conduct a physics-driven relaxation process described in Ref. [44, 45] to achieve this goal. In this particular pre-processing

method, the geometry surface is represented by a zero level-set contour within a signed distance level-set field $\phi(x, y, z)$:

$$\Gamma = \{(x, y, z)|\phi(x, y, z, t) = 0\}\,. \tag{5}$$

The negative and positive level-set values are for the inside and outside regions of the geometry, respectively.

After the definition of the geometry surface, a set of discrete particles is initially generated on a pre-defined lattice structure that is embedded within the geometry of interest. These particles are then transported to their respective isotropic and body-fitted positions by utilizing a constant background pressure field that is governed:

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{F}_p, \tag{6}$$

where $\mathbf{v}$ is the advection velocity, $\mathbf{F}_p$ represents the accelerations due to the background pressure. The right side of Eq. 6 can be written as SPH formulation:

$$\boldsymbol{F}_{p,i} = -\frac{2}{m_i}\sum_j V_i V_j p_0 \nabla_i W_{ij}, \tag{7}$$

where m is the particle mass, V the particle volume, p_0 the constant background pressure and $\nabla_i W_{ij}$ represents the gradient of the kernel function $W(|\mathbf{r}_{ij}|, h)$ with respect to particle i. Here, $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and h stands for the smoothing length. The time-step size Δt is constrained by the body force criterion:

$$\Delta t \leq 0.25\sqrt{\frac{h}{|\mathrm{d}\mathbf{v}/\mathrm{d}t|}}. \tag{8}$$

The position-updating method of particles is defined as

$$\mathbf{r}^{n+1} = \mathbf{r}^n + \mathrm{d}\mathbf{r} = \mathbf{r}^n + \frac{1}{2}\mathbf{F}_p^n \Delta t^2. \tag{9}$$

In addition, to achieve the body-fitted distribution, a boundary condition of either the bounding method in Ref. [44] or the 'static confinement' method in Ref. [45] should be imposed on those particles near the geometry surface. For more details, please check the references [44, 45].

2.3. Multi-phase Riemann-based SPH

To address the issue of density discontinuity across the ever-changing two-phase interface, we utilized a two-phase Riemann solver that incorporates pair-wise particle interaction to effectively deal with this phenomenon [38, 46]. Utilizing the two-phase Riemann SPH approach, the heavy phase serves as a moving boundary for the light phase, while simultaneously exhibiting characteristics of a free surface flow [38].

Following the reference [47, 38], the mass and momentum conservation equation of Eq.1 could be rewritten as the Riemann-based SPH discretization form:

$$\begin{cases} \frac{\mathrm{d}\rho_i}{\mathrm{d}t} = 2\rho_i \sum_j V_j(\mathbf{v}_i - \mathbf{v}^*)\nabla_i W_{ij} \\ \frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\frac{2}{m_i}\sum_j V_i V_j p^* \nabla_i W_{ij} + \frac{2}{m_i}\sum_j V_i V_j \frac{\eta_{ij}\mathbf{v_{ij}}}{r_{ij}}\frac{\partial W_{ij}}{\partial r_{ij}} + \mathbf{g} + \mathbf{f}_i^{s:p} + \mathbf{f}_i^{s:\nu} \end{cases}, \tag{10}$$

where V is particle volume, m is the particle mass, $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the relative velocity, $\mathbf{r}$ is the particle position, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the particle distance, $\nabla_i W_{ij} = \mathbf{e}_{ij}\frac{\partial W_{ij}}{\partial r_{ij}}$ is the gradient of the kernel function of particle i, and $\mathbf{v}^*$ and p^* are the solutions of inter-particle Riemann problem along the unit vector $\mathbf{e}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}}$[47].

The initial right and left states of the inter-particles Riemann problem can be reconstructed as

$$\begin{cases} (\rho_L, U_L, P_L) = (\rho_i, -\mathbf{v}_i \cdot \mathbf{e}_{ij}, p_i) \\ (\rho_R, U_R, P_R) = (\rho_j, -\mathbf{v}_j \cdot \mathbf{e}_{ij}, p_j) \end{cases}. \tag{11}$$

where the subscripts L and R are the left and right states, respectively [47]. Then, the Riemann solution with a dissipation limiter [47] could be achieved as:

$$\begin{cases} U^* = \bar{U} + \frac{p_L - p_R}{c(\rho_L + \rho_R)} \\ p^* = \bar{P} + \frac{\rho_L \rho_R \beta(U_L - U_R)}{\rho_L + \rho_R} \end{cases}, \tag{12}$$

where the dissipation limiter is $\beta = min[3max(U_L - U_R, 0), c^0]$. This limiter ensures that numerical dissipation is not applied to the fluid when it is subjected to an expansion wave, while it modulates the amount of numerical dissipation when the fluid is exposed to a compression wave[47]. Furthermore, we can obtain the value of $\mathbf{v}^*$ by the following equation:

$$\mathbf{v}^* = \left(U^* - \frac{\rho_L U_L + \rho_R U_R}{\rho_L + \rho_R}\right)\mathbf{e}_{ij} + \frac{\rho_i \mathbf{v}_i + \rho_j \mathbf{v}_j}{\rho_i + \rho_j}. \tag{13}$$

2.4. Density re-initialization

With the Riemann-based SPH method described in Section 2.3, the density field of the heavy phase can be treated as free-surface flow, and reinitialized at every advection time step [48] as:

$$\rho_i = \rho^0 \frac{\sum W_{ij}}{\sum W_{ij}^0} + max(0, \rho^* - \rho^0 \frac{\sum W_{ij}}{\sum W_{ij}^0})\frac{\rho^0}{\rho^*}. \tag{14}$$

where ρ^* stands for the density before reinitialization, and the superscript 0 refers to the initial states. Meanwhile, the density summation could be applied to the light phase which implies the light phase as an internal flow:

$$\rho_i = \rho^0 \frac{\sum W_{ij}}{\sum W_{ij}^0}. \tag{15}$$

2.5. Transport-velocity formulation

In this paper, we employ the formulation of the transport velocity in the light phase [49, 50] to avoid the unnatural void regions caused by the negative pressure during the simulation of fluid dynamics, as mentioned in[38].

$$\widetilde{\mathbf{v}}_i(t+\delta t) = \mathbf{v}_i(t) + \delta t(\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} - \frac{2p_b}{m_i}\sum_j V_i V_j \nabla_i W_{ij}). \tag{16}$$

where the particle advection velocity $\widetilde{\mathbf{v}}$ is used to advect the light phase particles while the $p_b = 4\rho^0 {c^0}^2$ represents a constant background pressure.

2.6. Boundary treatment

For boundary treatment, we employed dummy particles to implement the solid wall boundary condition [47, 34, 35] for two-phase fluid simulation. In principle, it treats the interaction between fluid particles and wall particles by solving a one-sided Riemann problem [47] along the normal direction of the wall. The left state of this one-side Riemann problem could be written as:

$$(\rho_L, U_L, P_L) = (\rho_f, -\mathbf{n}_w \cdot \mathbf{v}_f, P_f), \tag{17}$$

where the subscript f is for fluid particles, the $\mathbf{n}_w$ is the local wall normal direction. While the right states as follows:

$$\begin{cases} U_R = -U_L + 2u_w \\ P_R = P_L + \rho_f \mathbf{g} \cdot \mathbf{r}_{f_w} \end{cases}, \tag{18}$$

where the u_w is the velocity of the wall, the $\mathbf{r}_{f_w} = \mathbf{r}_w - \mathbf{r}_f$ and the density of the right state could be obtained by solving the artificial EoS. Therefore, the pressure force and the viscous force exerted by the wall on the fluid can be written as:

$$\begin{cases} \mathbf{f}_i^{s:p} = -\frac{2}{m_i}\sum_w V_i V_w p^* \nabla_i W_{iw} \\ \mathbf{f}_i^{s:\nu} = \frac{2}{m_i}\sum_w V_i V_w \frac{2\eta_i \eta_w}{\eta_i + \eta_w} \frac{\mathbf{v}_i - \mathbf{v}_w}{r_{iw}} \frac{\partial W_{iw}}{\partial r_{iw}} \end{cases}, \tag{19}$$

Here the subscript i is for fluid particles, and w is for wall particles. The intermediate pressure value p^* is obtained as:

$$p^* = \frac{\rho_f p_w + \rho_w p_f}{\rho_f + \rho_w}. \tag{20}$$

2.7. Bounding method for light phase

Massoud [38] successfully solved the appearance of non-physical void regions in violent two-phase flows with high-density ratios, which was observed in previous studies [51, 52]. However, it was critically observed that within such violent multiphase flow systems, characterized by substantial density disparities, the migration trajectory of light-phase particles that traverse through heavy-phase particles may result in interfacial penetration phenomena when reaching the

solid boundary [53]. To address this non-physical penetration issue, Massoud introduced a penalty force [53] to the light-phase particles when they are embedded within a cluster of heavy-phase particles while in contact with the wall to prevent penetration. Zhang [54] extended this method to the simulations of the flow-structure interaction and achieved promising results.

However, this parameter-dependent formulation exhibits significant operational limitations. The correction scheme necessitates frequent empirical parameter recalibration to accommodate varying physical configurations. Empirical observations reveal that under sufficiently refined spatial resolutions, even maximized repulsive penalty force fail to mitigate non-physical penetration artifacts of light-phase particulates. Moreover, even if the applied penalty force is sufficiently large to prevent particle penetration at the wall, calculating the forces acting on lightweight particles in this singular state significantly reduces the time step, ultimately hindering the normal progression of the simulation.

Typically, particles in this state can be referred to as singular particles. Once a particle enters this singularity, it can no longer be treated within the framework of a continuous medium, even if heavy particles surround it, rendering force calculations on such singular particles seemingly meaningless. Therefore, a transport-velocity-like methodology should be implemented to reposition particles exhibiting imminent or completed wall penetration. These particles are algorithmically relocated to the outermost position of the computational domain along the wall-normal direction while maintaining strict adherence to the original momentum conservation framework without introducing extraneous force terms.

Figure 1 shows the sketch of the Bounding method which is implemented in this paper. A corrective repositioning procedure is activated when light-phase particles (depicted in gray), entrained within the heavy-phase particle continuum (blue), penetrate or exceed wall boundaries. Specifically, this intervention triggers when the wall-normal distance d_1 of a light-phase particle satisfies $d_1 < 1/2\Delta x$ (where Δx denotes characteristic interparticle spacing and d_1 assumes negative values while the particle center completely crosses the boundary). Subsequently, the minimum normal distance d_2 between the subject particle and the first layer wall-adjacent particle center is calculated. The light-phase particle is then repositioned according to Equation 21.

$$\mathbf{r}_i = \mathbf{r}_i + \mathbf{d_2} \tag{21}$$

Here $\mathbf{d_2}$ is the vector form of the normal distance d_2, $\mathbf{r}_i$ is the particle position of the escaped one. This repositioning process preserves the acceleration and velocity of the particle, ensuring that it is placed correctly within the computational domain without introducing any additional influence on the momentum calculations of the heavy-phase particles.

In order to verify the effectiveness of the bounding method, we conduct a 2D two-phase dam-break flow for comparison with the previous penalty method. Fig. 2a illustrates a schematic of a dam-break problem. A two-dimensional rectangular tank of length $L = 5.366H$, where $H = 1$ denotes the characteristic

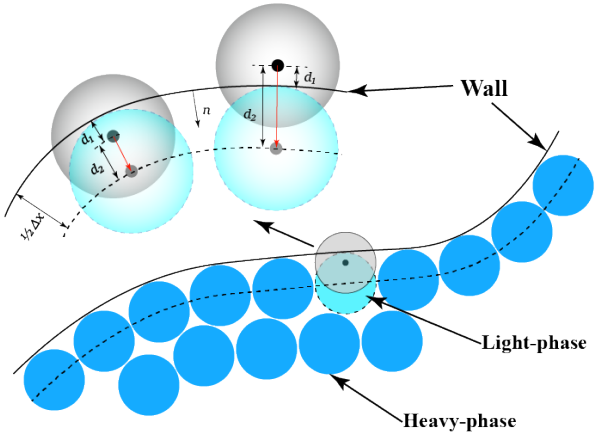


Figure 1: The sketch of the Bounding method for light-phase particle. Note that the blue circles represent heavy-phase particles near the wall, the gray circles represent light-phase particles that are about to or have penetrated the wall, and the cyan circles with dotted borders are where the escaped light-phase particles are sent back to the computation domain through the Bounding method. In the zoomed image, the dotted line represents the center position of the first layer particles near the wall. The n represents the wall normal, and Δx is the particle spacing. By calculating the distance d_1 from the center of the escaping particle to the wall and the distance d_2 to the center line of the first layer particles, the escaping particle is repositioned to the first layer close to the wall along the normal direction of the wall.

height, contains a water column of dimensions $2H \times H$ $(width \times height)$ immersed in an ambient air phase. The computational domain assumes inviscid flow conditions, with fluid densities prescribed as $\rho_w = 1$ (water) and $\rho_a = 0.001$ (air). Gravitational acceleration $g = 1$ acts in the negative y-direction, and all variables are presented in non-dimensional form. The simulation initiates with hydrostatic pressure-free water that undergoes instantaneous release at $t = 0$. Pressure monitoring is implemented via a transducer P, positioned on the downstream wall at a normalized vertical coordinate of $y/H = 0.2$ relative to the bottom boundary. The same relative resolution of $\Delta x = H/40$ is used for air and water in the simulation.

A comparative analysis of transient impact pressure profiles acquired from transducer P is presented in Fig.2b, contrasting the performance between penalty and bounding approaches. As anticipated, the pressure profile exhibits high-frequency fluctuations within a weakly compressible framework. However, the gray curve, which corresponds to the penalty method, demonstrates pronounced instability in oscillatory behavior beyond approximately $t = 8s$, correlated with interfacial penetration of light-phase particles through the wall as evidenced in Fig.2d. These nonphysical oscillations persist throughout the subsequent simulation timeline without observable attenuation or stabilization.

Figure 2c and Figure 2d compare the velocity and position distribution of air particles near the wall in the lower right corner of the computational domain when different methods are used at the same time step. It can be clearly

observed that the simulation with the penalty method has already seen light-phase particles penetrate the wall at this moment, and the maximum velocity is already an order of magnitude different from that of normal air particles. However, the simulation with the bounding method did not observe the phenomenon of particle penetration, and even the velocity of air particles that were already at the singularity did not present unreasonable values. In addition, since the penalty force is not only applied to particles that cross the wall, those light-phase particles close to the wall and entrained in the heavy-phase particles are also subject to penalty forces to varying degrees, and these unstable forces will be transmitted to other particles inside during the momentum calculation process, thus causing more particle velocities to present unreasonable values.

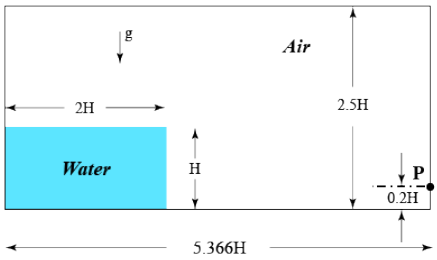
In fact, extensive testing has been conducted by us to evaluate these two approaches. Adjusting the magnitude of the penalty force based on specific conditions can mitigate or, in some cases, eliminate the particle penetration phenomenon. However, this penalty force method necessitates case-specific parameter tuning, making it less robust. In contrast, the bounding method effectively prevents light-phase particle penetration across all scenarios while ensuring computational stability. Therefore, in this study, the bounding method is adopted in place of the penalty force approach used by other researchers to control light-phase particle penetration.

2.8. Buoyancy-inspired Correction for Suppressing Numerical Gas Entrapment

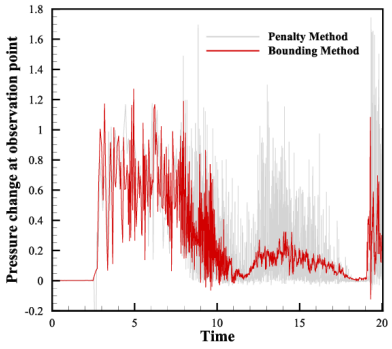
In long-time two-phase SPH simulations, we observed that a small number of gas particles may become entrained into the liquid phase and subsequently remain in liquid-dominated regions. Once this occurs, these particles may continue to drift downward instead of returning to the free surface. This behaviour deteriorates the sharpness of the gas–liquid interface and affects interface-based diagnostics. The phenomenon was first observed in extended dam-break simulations, as shown in Fig. 3a, and becomes more pronounced in the present tank-sloshing case, as shown in Fig. 5a, where repeated interface deformation makes gas entrapment and accumulation more likely.

Although this long-time gas-entrapment behaviour is rarely discussed explicitly in conventional short-duration dam-break benchmarks, it is consistent with the known sensitivity of multiphase SPH formulations to incomplete kernel support, large density ratios, and complex interface topology changes. Therefore, in the present work, a buoyancy-inspired local correction is introduced to suppress this numerical gas-entrapment effect.

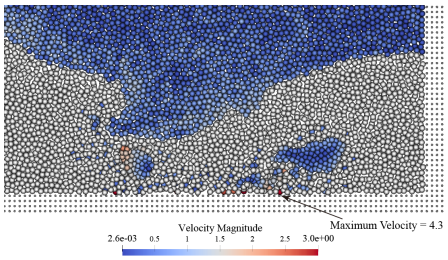
The correction is motivated by the physical picture of buoyancy-driven gas recovery in a liquid. In a real gas–liquid system, an entrapped gas bubble rises because the surrounding liquid pressure field produces a net upward restoring tendency, and the upward migration of the bubble is accompanied by the motion of the neighbouring liquid. The present correction mimics this behaviour at the particle level: an under-supported gas particle activates an upward correction on nearby liquid particles, and the resulting local gas–liquid configuration is transported toward the free surface through the standard two-phase contact dynamics. For this reason, the correction should be regarded as a localized



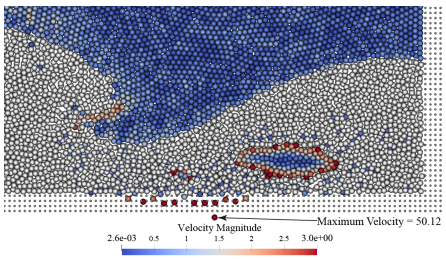
(a) Schematic illustration of the two-phase dam break problem



(b) comparison of the pressure signals recorded at sensor P

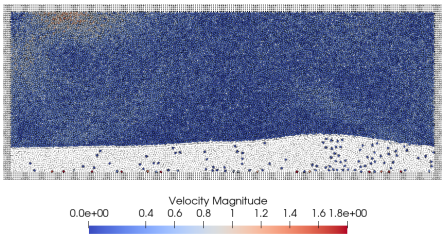


(c) Velocity distribution of light-phase particles with bounding method at time=8.2s

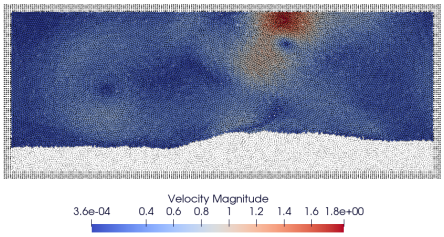


(d) Velocity distribution of light-phase particles with penalty method at time=8.2s

Figure 2: Comparison of the differences between the bounding method and the penalty method using a 2D two-phase dam-break case. Note that at about t=8s in sub-figure (b), the penalty method simulation begins to experience a dramatic pressure fluctuation, which corresponds to the light-phase particles penetrating the wall in sub-figure (d). The gray circles in sub-figure (c) and (d) are water particles, and the colored circles represent the velocity distribution of air particles.



(a) Velocity distribution of light-phase particles at $t = 40$ s without correction.



(b) Velocity distribution of light-phase particles at $t = 40$ s with correction.

Figure 3: Effect of the gas-entrapment correction in the extended dam-break case. (a) Without the correction, entrained gas particles remain isolated in the liquid bulk and gradually migrate downward. (b) With the correction, the local liquid-side upward restoring motion transports the entrapped gas–liquid configuration back toward the free surface.

stabilization term with buoyancy-like physical motivation rather than as a conservative interfacial-force model.

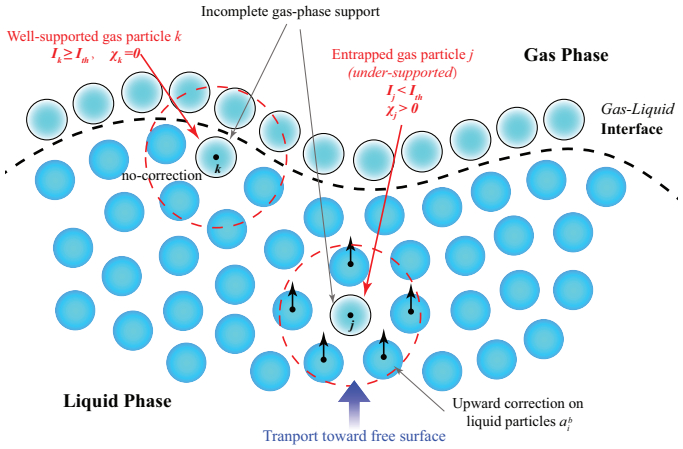


Figure 4: Schematic illustration of the buoyancy-inspired correction for suppressing numerical gas entrapment. A gas particle is not corrected solely because it geometrically enters the liquid region. If a gas particle, such as particle k, still retains sufficient gas-phase support, it satisfies $\mathcal{I}_k \geq \mathcal{I}_{\text{th}}$ and $\chi_k = 0$, and no correction is activated. In contrast, when an entrapped gas particle, such as particle j, becomes under-supported by gas-phase neighbours, it satisfies $\mathcal{I}_j < \mathcal{I}_{\text{th}}$ and $\chi_j > 0$. This activates an upward correction on the neighbouring liquid particles, rather than directly on the gas particle itself. The resulting upward liquid-side restoring motion transports the local gas–liquid configuration upward through the standard gas–liquid contact dynamics, helping the entrapped gas particle return toward the free surface.

Figure 4 illustrates the basic mechanism of the proposed correction. The correction is not triggered solely by geometric penetration of a gas particle into the liquid region; instead, it is activated only when the gas-phase support of that particle becomes insufficient. The corresponding mathematical definition is introduced below.

For each gas particle j, a gas-phase isolation indicator is first defined from its gas-phase neighbours as

$$\mathcal{I}_j = -\sum_{k \in \mathcal{N}_j^{aa}} \left(\frac{\partial W_{jk}}{\partial r_{jk}} \right) V_k r_{jk}, \tag{22}$$

where $\mathcal{N}_j^{aa}$ denotes the neighbour set associated with the air–air inner relation for gas particle j, V_k is the volume of gas neighbour k, and $r_{jk} = \|\mathbf{r}_j - \mathbf{r}_k\|$. For a well-supported gas particle in a nearly uniform gas region, $\mathcal{I}_j$ is of the order of the spatial dimension D. In contrast, when a gas particle enters a liquid-dominated region and loses most of its gas-phase neighbours, $\mathcal{I}_j$ becomes small. Therefore, $\mathcal{I}_j$ serves as an indicator of gas particles that are numerically isolated from the main gas phase.

A dimension-scaled threshold is introduced as

$$\mathcal{I}_{\text{th}} = \tau D, \tag{23}$$

where τ is a user-defined parameter. The corresponding activation weight is given by

$$\chi_j = \max\left(1 - \frac{\mathcal{I}_j}{\mathcal{I}_{\mathrm{th}}}, 0\right). \tag{24}$$

Accordingly, only gas particles satisfying $\mathcal{I}_j < \mathcal{I}_{\mathrm{th}}$ activate the correction. When a gas particle recovers sufficient gas-phase neighbours, $\mathcal{I}_j$ increases and χ_j automatically decreases to zero.

The correction is applied to neighbouring liquid particles rather than directly to the isolated gas particle. For a liquid particle i, the additional acceleration is assembled from its gas-contact neighbours as

$$\mathbf{a}_i^b = \sum_{j \in \mathcal{N}_i^{la}} W_{ij} V_i \left(-\chi_j C_b \mathbf{g}_{\mathrm{ref}}\right), \tag{25}$$

where $\mathcal{N}_i^{la}$ denotes the set of gas-phase neighbours of liquid particle i within the liquid–air contact relation, C_b is a dimensionless strength factor, and $\mathbf{g}_{\mathrm{ref}}$ is the reference downward gravitational acceleration used in the correction. Since $\mathbf{g}_{\mathrm{ref}}$ points downward, the term $-\chi_j C_b \mathbf{g}_{\mathrm{ref}}$ provides an upward correction to the liquid particles surrounding an under-supported gas particle. The corresponding correction force on liquid particle i is

$$\mathbf{F}_i^b = m_i \mathbf{a}_i^b. \tag{26}$$

Once activated, the correction biases the surrounding liquid particles upward rather than directly forcing the entrapped gas particle itself. Through the standard gas–liquid contact dynamics, this local liquid-side restoring motion transports the entrapped gas–liquid configuration back toward the free surface. As the gas particle recovers sufficient gas-phase support, χ_j automatically decreases to zero and the correction is deactivated.

The factor $W_{ij}V_i$ in Eq. (25) is used as a local contact weight in the present formulation. Since the correction is not constructed as an anti-symmetric pairwise force between gas and liquid particles, it is not strictly momentum-conservative. Its role is limited to suppressing numerical gas entrapment caused by incomplete gas-phase support, rather than modifying the resolved sloshing dynamics or replacing the standard two-phase momentum coupling.

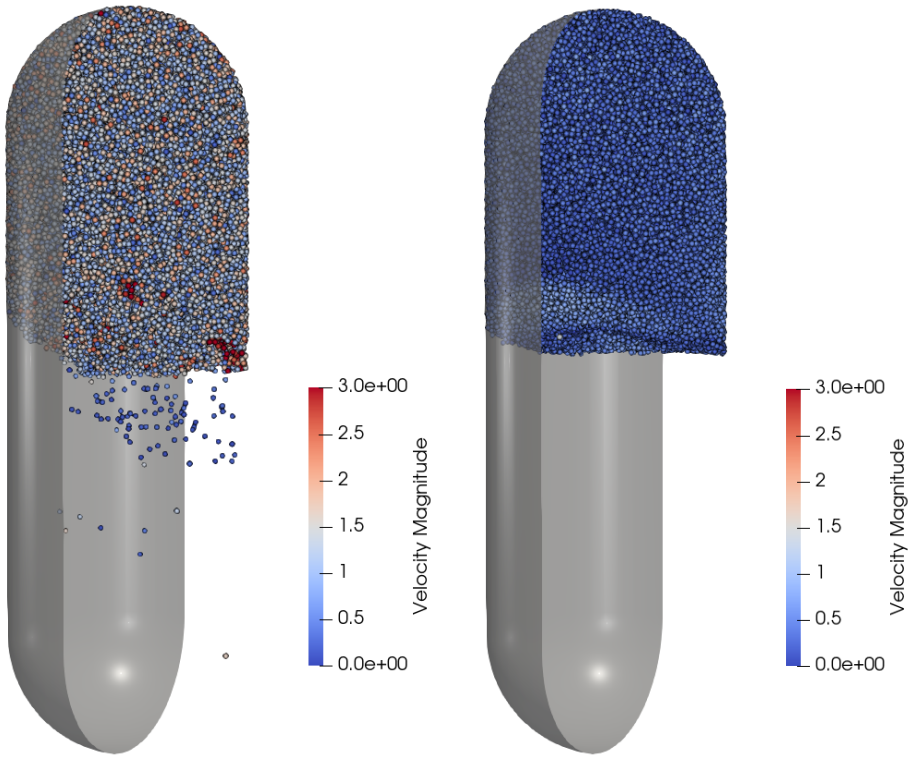
For the 3-D tank simulations reported in this work, the parameters are

$$\tau = 0.3, \qquad \mathcal{I}_{\mathrm{th}} = 0.3D = 0.9, \qquad C_b = 4.0.$$

With this correction, numerically entrained gas particles are prevented from persistently sinking into the liquid bulk, leading to a cleaner gas–liquid interface and more robust long-time heat-transfer diagnostics, as illustrated in Figs. 3b and 5b.

2.9. Two-phase thermal diffusion

Following Cleary [55], the thermal diffusion between SPH particles is evaluated in a pairwise conservative form. For particle i, the rate of change of internal



(a) Velocity distribution of gas particles at $t = 3.5$ s without correction.

(b) Velocity distribution of gas particles at $t = 3.5$ s with correction.

Figure 5: Effect of the gas-entrapment correction in the tank-sloshing case. (a) Without the correction, entrained gas particles remain suspended in the liquid bulk and continue to sink, while the disturbance is also transmitted into the upper gas layer. (b) With the correction, the entrained gas particles are transported back toward the gas–liquid interface, leading to a sharper and cleaner interface. The oscillation frequency is 1.5 Hz and the amplitude is 0.01 m.

energy due to thermal diffusion is written as

$$m_i \frac{\mathrm{d}U_i}{\mathrm{d}t} = \sum_j 2V_i V_j k_{ij}^H \left(T_i - T_j\right) \frac{r_{ij}}{r_{ij}^2 + \epsilon h_{ij}^2} \frac{\partial W_{ij}}{\partial r_{ij}}, \tag{27}$$

where m_i is the particle mass, $V_i = m_i/\rho_i$ is the particle volume, T_i is the particle temperature, r_{ij} is the distance between particles i and j, and W_{ij} is the smoothing kernel. The term h_{ij} denotes the smoothing length associated with the particle pair, and ϵ is a small regularization coefficient. In the present implementation, $\epsilon = 0.01$, so that the singular factor $1/r_{ij}$ is replaced by its regularized counterpart $r_{ij}/(r_{ij}^2 + \epsilon h^2)$.

The harmonic mean thermal conductivity is adopted for both single-phase and cross-material heat conduction:

$$k_{ij}^H = \frac{2k_i k_j}{k_i + k_j}. \tag{28}$$

This treatment gives the correct limiting behavior when two materials with different thermal conductivities are in contact.

Since the specific internal energy is related to temperature by $\mathrm{d}U_i = C_{p,i}\mathrm{d}T_i$, Eq. (27) can equivalently be written in temperature form as

$$\frac{\mathrm{d}T_i}{\mathrm{d}t} = \frac{1}{\rho_i C_{p,i}} \sum_j 2V_j k_{ij}^H \left(T_i - T_j\right) \frac{r_{ij}}{r_{ij}^2 + \epsilon h_{ij}^2} \frac{\partial W_{ij}}{\partial r_{ij}}. \tag{29}$$

Using $\alpha_i = k_i/(\rho_i C_{p,i})$, this expression may also be written as

$$\frac{\mathrm{d}T_i}{\mathrm{d}t} = \alpha_i \sum_j V_j \frac{4k_j}{k_i + k_j} \left(T_i - T_j\right) \frac{r_{ij}}{r_{ij}^2 + \epsilon h_{ij}^2} \frac{\partial W_{ij}}{\partial r_{ij}}. \tag{30}$$

Equations (27)–(30) are applied to both intra-phase and inter-phase particle pairs. For inter-phase thermal diffusion, the neighboring particle j is searched from the other phase within the kernel cut-off range.

2.10. Evaluation of interfacial heat transfer rate

The interfacial heat transfer rate across the liquid–gas interface is evaluated consistently with the pairwise thermal diffusion operator. At the continuum level, the heat transfer rate through an interface can be written in terms of Fourier's law as

$$\dot{Q}_\Gamma = \int_\Gamma k_\Gamma \frac{\Delta T}{\Delta n} \, \mathrm{d}A, \tag{31}$$

where k_Γ is the effective thermal conductivity at the interface, while Γ represents the interface between gas and liquid. $\Delta T/\Delta n$ is the normal temperature gradient across the interface, and $\mathrm{d}A$ is the interfacial area element.

In a discretized particle system, the concept of cross-section between two particles is less well-defined compared to the intersection between adjacent cells in a grid-based spatial framework. However, Wang [56] proposed a method for

computing the cross-sectional area between particles by analyzing flux calculations in the Eulerian formulation of Smoothed Particle Hydrodynamics (SPH) and the finite volume method (FVM). He analyzed the governing equations in their conservation form and highlighted that the interface area vector, which is required for flux calculation along the interface normal direction, is expressed as:

$$\mathbf{A_i} = \begin{cases} \mathbf{A}_{ij}^{SPH} = |A_{ij}|\mathbf{n}_{ij}^{SPH} = 2V_iV_j\nabla W_{ij} = 2V_iV_j\frac{\partial W_{ij}}{\partial r_{ij}}\mathbf{e}_{ij}^{SPH}, & \text{In SPH Form} \\ \mathbf{A}_{ij}^{FVM} = |A_{ij}|\mathbf{n}_{ij}^{FVM} = S_{ij}\mathbf{n}_{ij}^{FVM}, & \text{In FVM Form} \end{cases} \tag{32}$$

where S_{ij} represents the cross-sectional area between cell i and cell j in FVM framework or particle i and particle j in SPH framework, while the unit normal vector at the interface is denoted as $\mathbf{n}_{ij}^{FVM}$ or $\mathbf{n}_{ij}^{SPH}$ correspondingly, V_i and V_j are the volumes of particle i and j, respectively. Thus, the SPH pairwise interface area vector can be expressed as

$$\mathbf{A}_{ij}^{SPH} = 2V_iV_j\frac{\partial W_{ij}}{\partial r_{ij}}\mathbf{e}_{ij}^{SPH} \tag{33}$$

where $\mathbf{e}_{ij}$ is the unit vector along the particle pair. Since $\partial W_{ij}/\partial r_{ij}$ is negative inside the kernel support for standard smoothing kernels, we define

$$\omega_{ij} = -\frac{\partial W_{ij}}{\partial r_{ij}} \geq 0. \tag{34}$$

The positive projected pairwise interface area is therefore

$$A_{ij}^{SPH,+} = 2V_iV_j\omega_{ij}. \tag{35}$$

For a liquid particle $i \in \Omega_L$ and a gas particle $j \in \Omega_G$, the temperature gradient normal to the interface is approximated by the same regularized pairwise gradient used in the thermal diffusion operator,

$$\left(\frac{\Delta T}{\Delta n}\right)_{ij} = (T_j - T_i)\,\frac{r_{ij}}{r_{ij}^2 + \epsilon h^2}. \tag{36}$$

Substituting Eqs. (28), (35), and (36) into the discrete form of Eq. (31), the pairwise heat transfer rate from the gas phase to the liquid phase becomes

$$\dot{Q}_{ij}^{G\to L} = k_{ij}^H\,(T_j - T_i)\,A_{ij}^{SPH,+}\,\frac{r_{ij}}{r_{ij}^2 + \epsilon h^2}. \tag{37}$$

Equivalently, using Eq. (35),

$$\dot{Q}_{ij}^{G\to L} = 2V_iV_jk_{ij}^H\,(T_j - T_i)\,\omega_{ij}\,\frac{r_{ij}}{r_{ij}^2 + \epsilon h^2}. \tag{38}$$

The sign convention is such that $\dot{Q}_{ij}^{G\to L} > 0$ indicates heat transfer from the gas phase to the liquid phase.

The total instantaneous interfacial heat transfer rate is obtained by summing over all liquid–gas contact particle pairs,

$$\dot{Q}_{\Gamma}^{G\to L} = \sum_{i\in\Omega_L}\sum_{j\in\Omega_G} 2V_iV_jk_{ij}^{H}\left(T_j - T_i\right)\omega_{ij}\frac{r_{ij}}{r_{ij}^2+\epsilon h^2}. \tag{39}$$

This formulation is the cross-phase contribution of Eq. (27) evaluated with a prescribed direction of heat transfer. It is therefore consistent with the thermal diffusion operator used to evolve the temperature field.

The physical dimension of Eq. (39) is also consistent with a heat transfer rate. In three dimensions, $V_iV_j\omega_{ij}$ has the dimension of area because $[V_iV_j] = \mathrm{m}^6$ and $[\omega_{ij}] = \mathrm{m}^{-4}$, giving $[A_{ij}^{SPH,+}] = \mathrm{m}^2$. Hence,

$$\left[k_{ij}^{H}\left(T_j - T_i\right)A_{ij}^{SPH,+}\frac{r_{ij}}{r_{ij}^2+\epsilon h^2}\right] = \frac{\mathrm{W}}{\mathrm{m\,K}}\cdot\mathrm{K}\cdot\mathrm{m}^2\cdot\frac{1}{\mathrm{m}} = \mathrm{W}. \tag{40}$$

The summation in Eq. (39) therefore gives the instantaneous interfacial heat transfer rate in units of power.

The accumulated interfacial heat transfer from the gas phase to the liquid phase is obtained by integrating the instantaneous heat transfer rate over time,

$$E_{\Gamma}^{G\to L}(t) = \int_0^t \dot{Q}_{\Gamma}^{G\to L}(\tau)\,\mathrm{d}\tau. \tag{41}$$

In the time-discrete simulation, this quantity is accumulated as

$$E_{\Gamma}^{G\to L,n+1} = E_{\Gamma}^{G\to L,n} + \Delta t^n\dot{Q}_{\Gamma}^{G\to L,n}, \tag{42}$$

where Δt^n is the physical time step. When the accumulated heat transfer is evaluated from recorded output data, a trapezoidal integration may be used:

$$E_{\Gamma}^{G\to L}(t_N) \approx \sum_{n=0}^{N-1}\frac{\dot{Q}_{\Gamma}^{G\to L,n} + \dot{Q}_{\Gamma}^{G\to L,n+1}}{2}\left(t_{n+1} - t_n\right). \tag{43}$$

In addition to the instantaneous heat-transfer rate, the same pairwise flux measure can be used to define an effective interfacial area associated with the normal heat-transfer budget. The scalar quantity $A_{ij}^{SPH,+}$ in Eq. (35) is the magnitude of the pairwise flux measure. Directly summing $A_{ij}^{SPH,+}$ over all liquid–gas contact pairs, however, does not generally give a geometric interface area, because oblique and nearly tangential pairwise measures are also accumulated as positive contributions. Since Eq. (31) represents a normal flux through the interface, only the component of the pairwise flux measure projected onto the local interface normal should contribute to the effective area associated with interfacial heat transfer.

Accordingly, the effective interfacial area is defined as

$$A_{\Gamma,\mathrm{eff}}(t) = \sum_{i\in\Omega_L}\sum_{j\in\Omega_G(i)} A_{ij}^{SPH,+}\left|\mathbf{e}_{ij}^{SPH}\cdot\mathbf{n}_{\Gamma,i}\right|, \tag{44}$$

where $\Omega_G(i)$ denotes the gas-phase contact neighbors of liquid particle i, and $\mathbf{n}_{\Gamma,i}$ is the local unit normal pointing from the liquid phase to the gas phase. The absolute value is used because $A_{\Gamma,\mathrm{eff}}$ measures the magnitude of the interface area participating in the normal flux budget, whereas the sign of the heat transfer is determined by the temperature difference in Eq. (38).

When the interface geometry is known, $\mathbf{n}_{\Gamma,i}$ can be prescribed directly. For a deforming liquid–gas interface, it is estimated from the weighted contact directions as

$$\mathbf{n}_{\Gamma,i} = \frac{\displaystyle\sum_{j\in\Omega_G(i)} V_j \omega_{ij} \mathbf{e}_{ij}^{SPH}}{\left| \displaystyle\sum_{j\in\Omega_G(i)} V_j \omega_{ij} \mathbf{e}_{ij}^{SPH} \right|}. \tag{45}$$

The sign of this normal does not affect Eq. (44), because only the projected magnitude is used. Therefore, $A_{\Gamma,\mathrm{eff}}$ should be interpreted as an operator-consistent effective interfacial area for the normal heat-transfer budget, rather than as a purely geometrically reconstructed surface area.

2.11. Time integration

In this work, the dual-criteria time-stepping method [48] is adopted to improve computational efficiency. With this strategy, the particle-neighbour list and the associated kernel quantities are updated on the advection time scale, whereas the hydrodynamic variables are advanced on the acoustic time scale.

The advection time step Δt_{ad} and the acoustic time step Δt_{ac} are defined as

$$\begin{cases} \Delta t_{ad} = CFL_{ad} \min\left(\dfrac{h}{|\mathbf{v}|_{\max}}, \dfrac{h^2}{\nu} \right), \\ \Delta t_{ac} = CFL_{ac} \dfrac{h}{c + |\mathbf{v}|_{\max}}, \end{cases} \tag{46}$$

where $CFL_{ad} = 0.25$ and $CFL_{ac} = 0.6$, h is the smoothing length, $|\mathbf{v}|_{\max}$ is the maximum particle advection velocity, ν is the kinematic viscosity, and c is the artificial sound speed.

For the two-phase flow, the actual advection and acoustic time steps are taken as the minima over the two phases:

$$\begin{cases} \Delta t_{ad} = \min\left(\Delta t_{ad}^{l}, \Delta t_{ad}^{h} \right), \\ \Delta t_{ac} = \min\left(\Delta t_{ac}^{l}, \Delta t_{ac}^{h} \right), \end{cases} \tag{47}$$

where the superscripts l and h denote the light and heavy phases, respectively.

Within one advection step Δt_{ad}, the pressure relaxation is carried out by several acoustic substeps Δt_{ac} using the position-based Verlet scheme [57]. In each acoustic step, the density and position are first updated to the midpoint:

$$\begin{cases} \rho_i^{n+\frac{1}{2}} = \rho_i^n + \dfrac{1}{2} \Delta t_{ac} \dfrac{\mathrm{d}\rho_i}{\mathrm{d}t}, \\ \mathbf{r}_i^{n+\frac{1}{2}} = \mathbf{r}_i^n + \dfrac{1}{2} \Delta t_{ac} \mathbf{v}_i^n, \end{cases} \tag{48}$$

followed by the velocity update

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \Delta t_{ac} \frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t}, \tag{49}$$

and finally the second half-step update of position and density:

$$\begin{cases} \mathbf{r}_i^{n+1} = \mathbf{r}_i^{n+\frac{1}{2}} + \dfrac{1}{2}\Delta t_{ac}\mathbf{v}_i^{n+1}, \\ \rho_i^{n+1} = \rho_i^{n+\frac{1}{2}} + \dfrac{1}{2}\Delta t_{ac}\dfrac{\mathrm{d}\rho_i}{\mathrm{d}t}. \end{cases} \tag{50}$$

For the coupled two-phase heat-transfer problem, thermal diffusion is advanced with an additional thermal substep. For the explicit integration of the thermal diffusion equation, the thermal time step for each phase is taken as

$$\Delta t_{th}^m = 0.5\frac{\rho_m c_{p,m} h^2}{k_m}, \qquad m \in \{l, h\}, \tag{51}$$

where k_m is the thermal conductivity, ρ_m is the density, and $c_{p,m}$ is the specific heat capacity of phase m. The actual thermal substep is then chosen as

$$\Delta t_{th} = \min\left(\Delta t_{th}^l, \Delta t_{th}^h\right). \tag{52}$$

Within each acoustic step Δt_{ac}, the thermal relaxation is sub-cycled using Δt_{th} until the full acoustic interval is consumed, i.e.

$$\sum_{s=1}^{N_{th}} \delta t_{th}^{(s)} = \Delta t_{ac}, \qquad \delta t_{th}^{(s)} \le \Delta t_{th}, \tag{53}$$

where N_{th} is the number of thermal substeps within one acoustic step. Accordingly, the flow field is advanced by the dual-criteria time-stepping strategy, whereas the temperature field is updated by thermal diffusion sub-iterations inside each acoustic step.

3. Validation and Resolution-Convergence Analysis

In this section, the proposed numerical methodology is assessed through a series of validation and resolution-convergence studies. The hydrodynamic performance of the two-phase solver is first examined using a violently sloshing flow in a laterally excited vertical cylindrical tank, by comparison with both experimental measurements and VOF simulations reported in the literature. The thermal diffusion model is then verified using heat conduction in a composite slab, for which analytical reference solutions are available. Subsequently, the proposed interfacial heat-transfer estimator is validated using both the composite-slab problem and a quiescent three-dimensional cryogenic tank with an initially flat liquid–gas interface.

These validation tests are required because the present study focuses not only on the global sloshing response, but also on the heat transferred across the gas–liquid interface. Therefore, the validation is designed to examine three related aspects of the numerical method: the prediction of free-surface sloshing dynamics, the accuracy of thermal diffusion across material interfaces, and the quantitative evaluation of interfacial heat transfer. In addition, resolution-convergence analyses are carried out for the free-surface response and for the interfacial heat-transfer statistics, so that the particle resolution used in the subsequent cryogenic-tank simulations can be selected on a quantitative basis.

3.1. Hydrodynamic validation: two-phase sloshing in a vertical cylindrical tank under lateral excitation

To assess the capability of the present two-phase SPH formulation in predicting severe sloshing, we consider the laterally excited cylindrical-tank benchmark reported in Ref. [58]. In that study, experimental measurements and VOF simulations performed with OpenFOAM were both provided, which makes this case suitable for simultaneous validation against experiments and against a well-established Eulerian reference method. As in the reference configuration, the tank is subjected to a horizontal sinusoidal excitation along the x-direction.

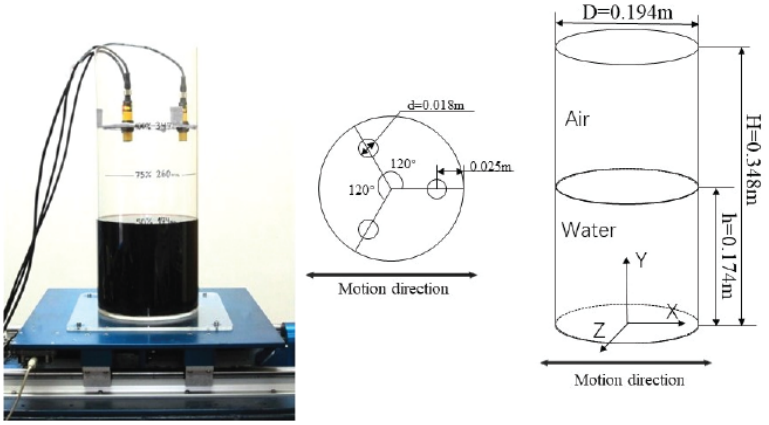


Figure 6: Experimental layout[58], tank geometry and sensor locations

The tank geometry and the locations of the sensors are shown in Fig. 6. Three ultrasonic sensors are mounted above the tank to measure the free-surface evolution at the circumferential positions 0°, 120°, and 240°, respectively. Each sensing region has a diameter of 0.018 m, and the distance from its center to the tank wall is 0.025 m in order to avoid interference. The imposed lateral acceleration is

$$a_x(t) = -4\pi^2 f^2 A \sin(2\pi f t), \tag{54}$$

where f is the excitation frequency, A is the excitation amplitude, and t is time.

The first-mode natural frequency of the liquid in the cylindrical tank can be estimated analytically as [59]

$$f_n = \frac{1}{2\pi}\left[\frac{1.84g}{R}\tanh\left(\frac{1.84h}{R}\right)\right]^{1/2},$$

where g is the gravitational acceleration, R is the tank radius, and h is the filling level. For the present filling level of 174 mm, Ref. [58] reported $f_n \approx 2.17$ Hz.

Two experimental conditions from Ref. [58] are considered here. Case A has an excitation frequency away from resonance and is used to validate the time histories of the free-surface elevation. Case B is close to the first-mode natural frequency and is used to assess the predicted wave shapes under strongly nonlinear sloshing. The corresponding case parameters are listed in Table 1, and the main physical properties are summarized in Table 2.

Case	Filling level (mm)	Imposed amplitude (mm)	Imposed frequency (Hz)
A	174	7.5	1.63
B	174	7.5	2.22

Table 1: Parameters of the validation cases

For the two-phase flow case, we set the same water and air sound speeds according to Ref.[38, 54]. The characteristic maximum velocity is estimated as $2\sqrt{gH_w}$, where H_w is the initial water depth, and the numerical sound speed is set to $20\sqrt{gH_w}$. The smoothing lengths of the fluid and solid particles are taken as $h_f = 1.3\Delta x$ and $h_s = 1.15\Delta x$, respectively, where Δx is the initial particle spacing used for water, air, and the tank wall. The free-surface elevation measured at each sensor location is defined as

$$\eta = h_{wave} - h_w, \tag{55}$$

where h_{wave} is the instantaneous local wave height and h_w is the initial water depth.

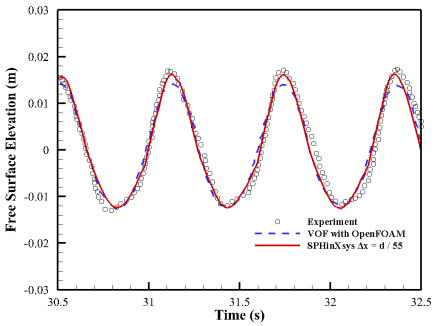
Properties	Values
Density of water ρ_w $[kg/m^3]$	1000.0
Density of air ρ_a $[kg/m^3]$	1.226
Gravitational acceleration g $[m/s^2]$	9.81
Height of the tank H $[m]$	0.348
Internal diameter of the tank D $[m]$	0.194
Water depth H_w $[m]$	0.174

Table 2: Physical parameters of the water, air, and tank

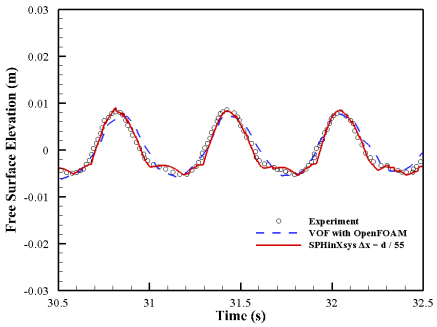
Figure 7 compares the free-surface elevations measured in the experiments [58], the VOF results from OpenFOAM reported in the same reference, and the

present SPHinXsys results [34, 35]. In the experiments, approximately 30 s were required before a periodic steady regime was reached; therefore, the time axis in Fig. 7 starts after 30 s. The present SPHinXsys simulation uses an initial particle spacing of $\Delta x = D/65 \approx 0.003$ m, while the OpenFOAM simulation in Ref. [58] used a mesh size of 0.0025 m.

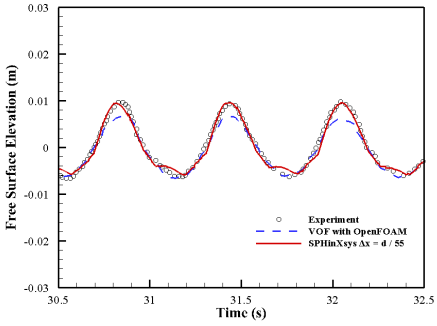
Overall, the present results show better agreement with the experimental measurements than the VOF results in both amplitude and phase. From Fig. 7a to Fig. 7c, the SPHinXsys solution reproduces the absolute wave heights more accurately and better captures the local nonsmooth trough evolution observed at S2 and S3. These features are associated with strongly nonlinear free-surface dynamics and are less accurately represented by the OpenFOAM results reported in the reference.



(a) at the circumferential positions 0°



(b) at the circumferential positions 120°



(c) at the circumferential positions 240°

Figure 7: Free surface elevation of three detecting points. Note that the results from 'VOF Method' and 'Experiment' are from Ref.[58]

Figure 8 further compares the instantaneous wave profiles of the present method, the experiments, and the OpenFOAM solution for Case B. Since the excitation frequency in this case is close to the first natural frequency, the tank

undergoes violent sloshing with pronounced nonlinear wave deformation. Under this near-resonant condition, accurate prediction of the crest shape becomes particularly demanding.

It can be seen that the OpenFOAM solution captures the global oscillation but fails to reproduce several important details of the experimentally observed free surface, especially the sharp crest formation at the highest point of the wave. At some instants, the predicted orientation of the wave peak even differs from that observed in the experiment. By contrast, the present SPHinXsys results reproduce the experimentally observed crest topology and temporal evolution more faithfully. This comparison indicates that the proposed method is able to capture not only the global sloshing motion but also the local nonlinear free-surface structures that arise under severe excitation.

3.2. Thermal validation: heat conduction in a composite slab

To validate the heat-conduction model across material interfaces, we consider the composite-slab benchmark described in Refs. [60, 55]. Although the problem is represented by a two-dimensional particle distribution, the heat transfer is one-dimensional in the x-direction. The slab geometry is shown in Fig. 9, with a length of $x = 1.0$ and a width of $y = 0.5$. Two materials occupy the left and right halves of the domain, and the material interface is located at $x = 0.5$.

Three cases are considered in order to test the behavior of the method under different contrasts in specific heat capacity, thermal diffusivity, and density ratio. The corresponding nondimensional material properties are listed in Table 3. In all cases, the initial particle spacing is set to $\Delta x = 0.0125$, and the smoothing length is taken as $h = 1.15\Delta x$. The thermal conductivity values in Table 3 are chosen consistently with

$$\alpha = \frac{k}{\rho c_p}.$$

Properties	Values		
	Case 1	Case 2	Case 3
Density of left material ρ_l	1000.0	1000.0	1.226
Density of right material ρ_r	1000.0	2000.0	1000.0
Thermal conductivity of left material k_l	1.0	1.0	0.0254
Thermal conductivity of right material k_r	3.0	3.0	0.620
Specific heat capacity of left material c_p^l	1.0	1.0	1.012
Specific heat capacity of right material c_p^r	3.0	1.0	4.179
Thermal diffusivity of left material α_l	0.001	0.001	0.02047
Thermal diffusivity of right material α_r	0.001	0.0015	0.000148

Table 3: Thermal properties of the left and right materials

Figures 10 and 11 show the temperature distributions across the slab for Cases 1–3 at representative times. In all cases, the present SPHinXsys results agree closely with the analytical solutions. The method correctly reproduces

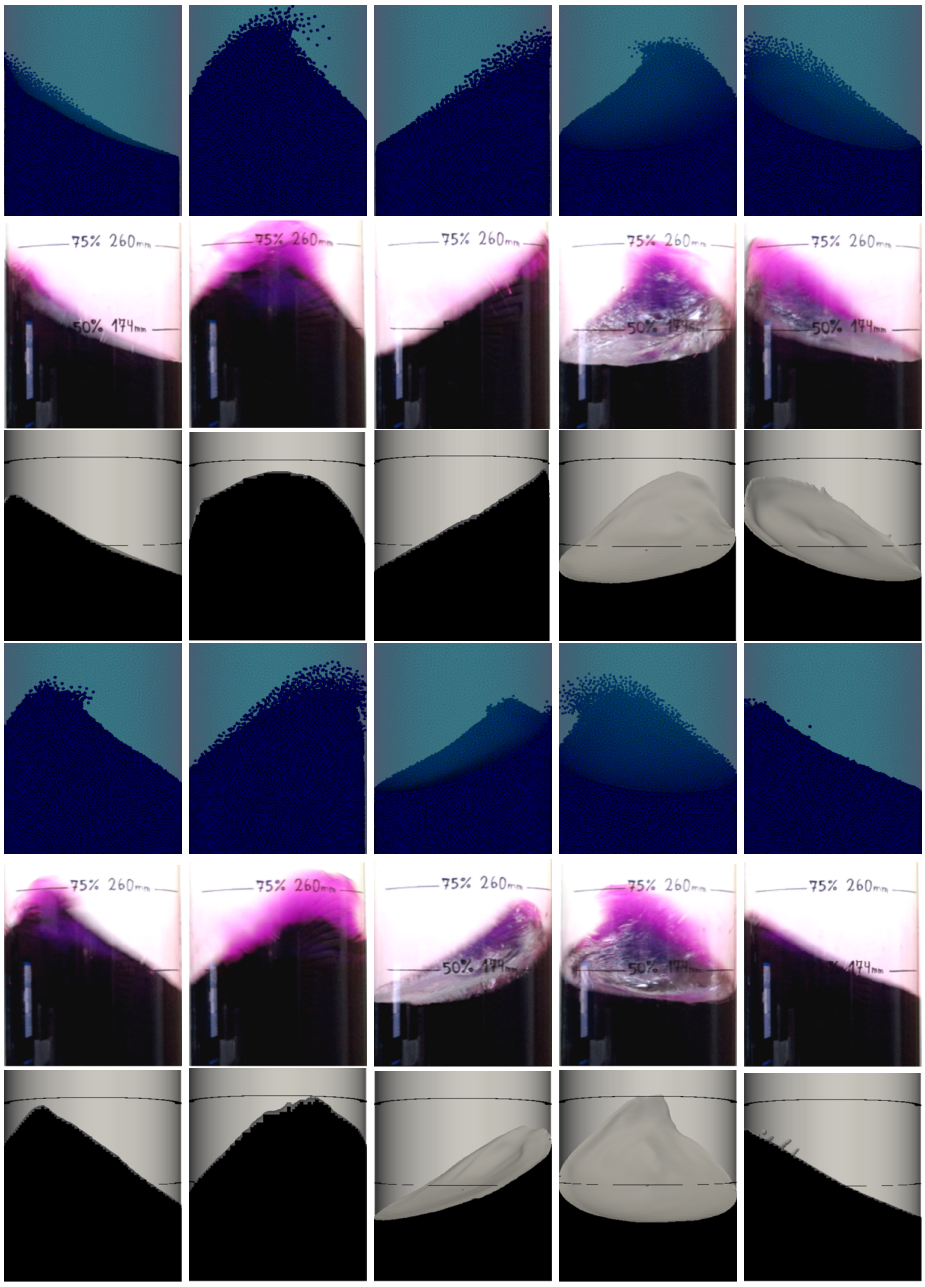


Figure 8: Wave profiles of SPHinXsys simulation (The first and fourth rows), the experiment (The second and fifth rows)[58], and VOF method based simulation (The third and sixth rows)[58] from 30.50s to 31.40s

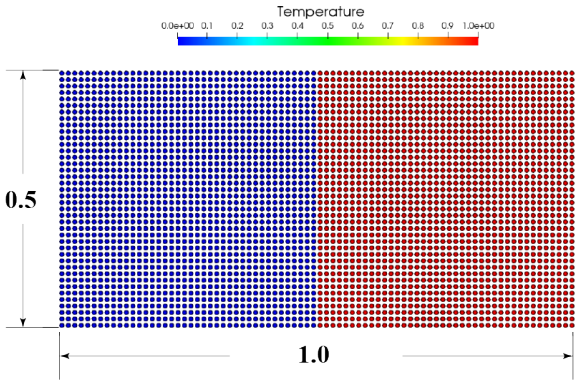
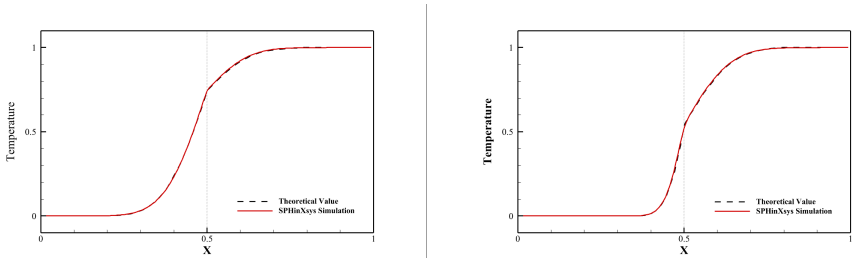


Figure 9: Initial temperature field in the composite slab

the continuous temperature field across the material interface together with the change in temperature gradient caused by the contrast in thermal properties.

For Case 3, where the density ratio between the two materials is particularly large, a slight deviation from the analytical solution can be observed near the interface at $t = 0.1$. This difference becomes much smaller at $t = 0.5$ and is further reduced under refinement. Such behavior is consistent with the kernel-smoothing effect inherent in SPH when sharp material contrasts are present. Overall, these results demonstrate that the proposed formulation can accurately predict heat conduction across interfaces between dissimilar materials and is sufficiently robust for the two-phase thermal problems considered in this work.

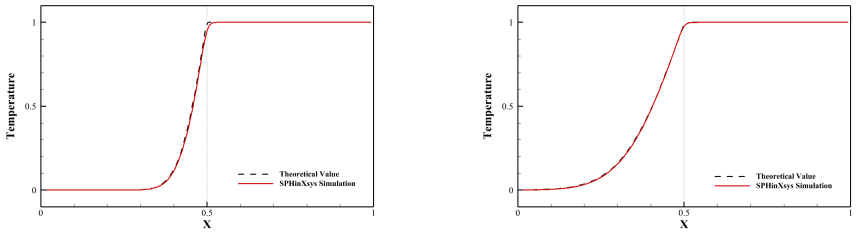


(a) (b)

Figure 10: Temperature profiles (a) Case 1: at t = 5.0, (b) Case 2: at t = 2.0

3.3. Interfacial heat-transfer validation in the 2D composite slab

The temperature-profile comparisons in the previous subsection verify that the present SPH formulation can accurately reproduce bulk heat diffusion in the composite slab. In addition to the temperature field itself, the present work also requires a quantitative estimate of the heat transferred across material interfaces. Therefore, the 2D composite-slab conduction problem (see Fig. 9) is further used to validate the proposed SPH interfacial heat-transfer estimator.



(a) (b)

Figure 11: Temperature profiles of case 3 (a) at t = 0.1; (b) at t = 0.5

The material interface is located at $x = 0.5$ and extends over $y \in [0, 0.5]$. A unit out-of-plane thickness is assumed, and therefore the sharp geometric interfacial area is

$$A_{\Gamma,\mathrm{geo}} = 0.5 \times 1 = 0.5 \text{ m}^2. \tag{56}$$

For the present interfacial heat-transfer validation, Case 3 in Table 3 is considered, corresponding to

$$\rho_l = 1.226, \quad c_{p,l} = 1.012, \quad k_l = 0.0254, \tag{57}$$

and

$$\rho_r = 1000.0, \quad c_{p,r} = 4.179, \quad k_r = 0.620. \tag{58}$$

The initial temperatures of the left and right materials are $T_l^0 = 0$ and $T_r^0 = 1$, respectively. The initial temperature difference is therefore $\Delta T = T_r^0 - T_l^0 = 1$.

The analytical reference is obtained from the transient thermal contact solution for two semi-infinite materials. The thermal effusivity of each material is defined as

$$e_i = \sqrt{k_i \rho_i c_{p,i}}, \tag{59}$$

and the effective effusivity controlling the interfacial heat flux is

$$e_{\mathrm{eq}} = \frac{e_l e_r}{e_l + e_r}. \tag{60}$$

Using the material properties of Case 3 gives

$$e_l = \sqrt{k_l \rho_l c_{p,l}} \approx 0.1775, \qquad e_r = \sqrt{k_r \rho_r c_{p,r}} \approx 50.9017, \tag{61}$$

and hence

$$e_{\mathrm{eq}} \approx 0.1769. \tag{62}$$

The reference interfacial heat-flux density is then

$$q''_{\Gamma,\mathrm{ref}}(t) = \frac{e_{\mathrm{eq}} \Delta T}{\sqrt{\pi t}}, \tag{63}$$

and the corresponding total interfacial heat-transfer rate is

$$\dot{Q}_{\Gamma,\mathrm{ref}}(t) = A_{\Gamma,\mathrm{geo}} q''_{\Gamma,\mathrm{ref}}(t). \tag{64}$$

At $t = 1\,\mathrm{s}$, this gives

$$\dot{Q}_{\Gamma,\mathrm{ref}}(1\,\mathrm{s}) \approx 0.0499\ \mathrm{W}. \tag{65}$$

The cumulative interfacial heat transfer is obtained by integrating the reference heat-transfer rate in time:

$$E_{\Gamma,\mathrm{ref}}(0,t) = \int_0^t \dot{Q}_{\Gamma,\mathrm{ref}}(t')\,\mathrm{d}t'. \tag{66}$$

Substituting the analytical heat-flux expression gives

$$E_{\Gamma,\mathrm{ref}}(0,t) = 2A_{\Gamma,\mathrm{geo}} e_{\mathrm{eq}} \Delta T \sqrt{\frac{t}{\pi}}. \tag{67}$$

Therefore, over $0 \sim 5\,\mathrm{s}$,

$$E_{\Gamma,\mathrm{ref}}(0, 5\,\mathrm{s}) \approx 0.223\ \mathrm{W \cdot s}. \tag{68}$$

Table 4 compares the SPH-predicted instantaneous interfacial heat-transfer rate at $t \approx 1\,\mathrm{s}$ and the cumulative interfacial heat transfer over $0 \sim 5\,\mathrm{s}$ with the analytical reference values.

Table 4: **Interfacial heat-transfer validation in the 2D composite slab.** Reference values: $\dot{Q}_{\Gamma,\mathrm{ref}}(1\,\mathrm{s}) = 0.0499\,\mathrm{W}$ and $E_{\Gamma,\mathrm{ref}}(0, 5\,\mathrm{s}) = 0.223\,\mathrm{W \cdot s}$.

dp (m)	$\dot{Q}_\Gamma^{\mathrm{SPH}}$ (1 s) (W)	Error in $\dot{Q}$ (%)	E_Γ^{SPH} (0, 5 s) (W·s)	Error in E (%)
0.05	0.04829	-3.23	0.215	-3.59
0.025	0.04857	-2.67	0.217	-2.69
0.0125	0.04867	-2.46	0.218	-2.24
0.00625	0.04874	-2.32	0.219	-1.79
0.003125	0.04880	-2.20	0.220	-1.35

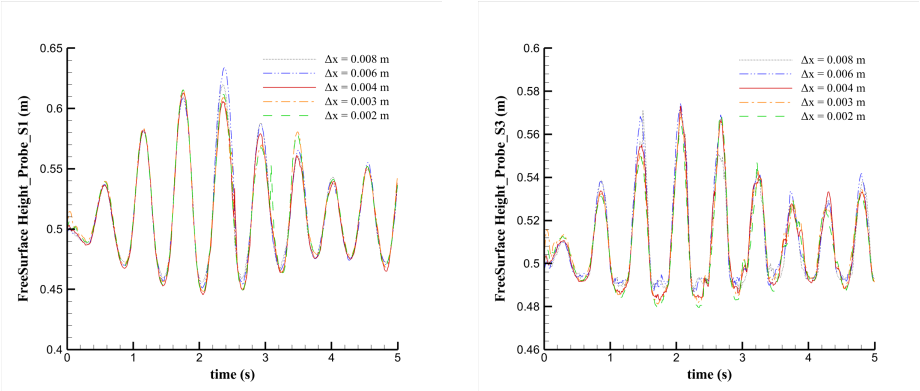
The instantaneous interfacial heat-transfer rate predicted by the SPH estimator shows weak sensitivity to particle resolution. As dp decreases from 0.05 to $0.003125\,\mathrm{m}$, $\dot{Q}_\Gamma^{\mathrm{SPH}}(1\,\mathrm{s})$ varies only from 0.04829 to $0.04880\,\mathrm{W}$. The corresponding relative deviation from the analytical reference decreases from 3.23% to 2.20%. The cumulative interfacial heat transfer also improves with refinement. $E_\Gamma^{\mathrm{SPH}}(0, 5\,\mathrm{s})$ increases from 0.215 to $0.220\,\mathrm{W \cdot s}$, and the deviation from the analytical reference decreases from 3.59% to 1.35%.

Overall, this benchmark demonstrates that the proposed SPH interfacial heat-transfer estimator provides a stable and quantitatively accurate prediction of the heat transferred across a material interface. Together with the temperature-profile validation in the previous subsection, this result confirms that the present formulation can reproduce both the bulk thermal diffusion and the associated interfacial heat-transfer rate in a composite conduction problem.

3.4. Resolution convergence of free-surface dynamics in the cryogenic tank

To further examine the resolution dependence of the free-surface dynamics in the tank configuration used in the subsequent application study, an additional mesh-independence check was carried out using the realistic cryogenic tank. The tank geometry and the initial fill level are shown in Fig. 13, and the corresponding thermophysical properties are listed in Table 6. The most severe lateral-excitation condition considered in this work was selected for this test, namely the maximum imposed lateral acceleration of $\Gamma = 0.10g$ and an excitation frequency of $f = 1.8\,\mathrm{Hz}$, which is close to the first sloshing natural frequency of the tank.

Figure 12 compares the time histories of the free-surface elevation recorded at probes S1 and S3 for different particle resolutions. At coarse resolutions, noticeable differences can still be observed in the peak amplitudes and in the local phase of the liquid-level oscillation. As the particle spacing is reduced, the predicted free-surface responses become increasingly consistent. In particular, the curves obtained with $dp = 0.004\,\mathrm{m}$ and finer resolutions almost overlap at both probes, indicating that the dominant sloshing period, phase evolution, and wave-amplitude response are already only weakly affected by further refinement from this level onward.



(a) (b)

Figure 12: Resolution dependence of the free-surface elevation histories at probes S1 (a) and S3 (b) under the strongest lateral excitation condition, $\Gamma = 0.10g$ and $f = 1.8$ Hz.

3.5. Interfacial heat-transfer validation and resolution convergence in a static three-dimensional tank

To validate the proposed interfacial heat-transfer statistic in a three-dimensional gas–liquid configuration and to quantify its resolution requirement, we consider a quiescent cryogenic tank containing liquid methane and gaseous nitrogen. The tank has the same geometry and initial fill level as the realistic cryogenic-tank configuration used in the subsequent application study, as shown in Fig. 13. The corresponding thermophysical properties are listed in Table 6.

The initial liquid–gas interface is flat, and no external sloshing excitation is imposed. Fluid motion is therefore suppressed, so that the heat transfer across the interface is governed primarily by thermal diffusion. This setting provides a controlled three-dimensional test for the interfacial heat-transfer estimator: the numerical interfacial heat-transfer rate and the cumulative transferred heat can be compared with a one-dimensional transient conduction reference, while the same case can also be used to examine the convergence of these quantities with particle resolution.

The reference solution is constructed by approximating the initially flat liquid–gas interface as a perfect thermal contact between two semi-infinite media. For a semi-infinite body initially at temperature T_i and suddenly exposed to a boundary temperature T_s, the classical transient conduction solution gives the surface heat-flux density as

$$q''(t) = k\frac{T_s - T_i}{\sqrt{\pi\alpha t}} = b\,\frac{T_s - T_i}{\sqrt{\pi t}}, \tag{69}$$

where

$$\alpha = \frac{k}{\rho c_p}, \qquad b = \sqrt{k\rho c_p} \tag{70}$$

are the thermal diffusivity and thermal effusivity, respectively.

Applying Eq. (69) to the liquid and gas sides of the interface gives

$$q''_{\Gamma,L}(t) = b_L\frac{T_\Gamma - T_{L,0}}{\sqrt{\pi t}}, \tag{71}$$

and

$$q''_{\Gamma,G}(t) = b_G\frac{T_{G,0} - T_\Gamma}{\sqrt{\pi t}}, \tag{72}$$

where T_Γ is the interface temperature, and $T_{L,0}$ and $T_{G,0}$ are the initial temperatures of the liquid and gas phases. For perfect thermal contact, the heat flux must be continuous across the interface, i.e.

$$q''_{\Gamma,L}(t) = q''_{\Gamma,G}(t), \tag{73}$$

which yields

$$b_L\left(T_\Gamma - T_{L,0}\right) = b_G\left(T_{G,0} - T_\Gamma\right). \tag{74}$$

Solving for T_Γ gives

$$T_{\Gamma,\mathrm{ref}} = \frac{b_L T_{L,0} + b_G T_{G,0}}{b_L + b_G}. \tag{75}$$

Substituting Eq. (75) back into the surface-flux relation leads to the reference interfacial heat-flux density

$$q''_{\Gamma,\mathrm{ref}}(t) = \frac{b_L b_G}{b_L + b_G}\,\frac{|T_{L,0} - T_{G,0}|}{\sqrt{\pi t}}. \tag{76}$$

According to Fig. 13, the inner diameter of the tank is $D = 0.288\,\mathrm{m}$, and the initial liquid surface lies in the cylindrical section of the tank. Therefore, the initial sharp interfacial area is

$$A_{\Gamma,\mathrm{geo}} = \frac{\pi D^2}{4}. \tag{77}$$

The corresponding total reference interfacial heat-transfer rate is

$$\dot{Q}_{\Gamma,\mathrm{ref}}(t) = A_{\Gamma,\mathrm{geo}}\, q''_{\Gamma,\mathrm{ref}}(t), \tag{78}$$

and the cumulative interfacial heat transfer is

$$E_{\Gamma,\mathrm{ref}}(0,t) = \int_0^t \dot{Q}_{\Gamma,\mathrm{ref}}(t')\,\mathrm{d}t'. \tag{79}$$

Using the material properties in Table 6 and the tank geometry in Fig. 13, the reference values at $t = 3.4\,\mathrm{s}$ are

$$\dot{Q}_{\Gamma,\mathrm{ref}}(3.4\,\mathrm{s}) \approx 2.31\ \mathrm{W}, \tag{80}$$

and

$$E_{\Gamma,\mathrm{ref}}(0, 3.4\,\mathrm{s}) \approx 15.71\ \mathrm{J}. \tag{81}$$

In the numerical simulations, the last sampled output is located at $t_{\mathrm{end}} \approx 3.40$–$3.42\,\mathrm{s}$, so slight fluctuations around the theoretical value at $t = 3.4\,\mathrm{s}$ are expected.

Table 5 summarizes the numerical results obtained at different resolutions for the end-time interfacial heat-transfer rate, the cumulative interfacial heat transfer, and the effective interfacial area in the static tank. Only the liquid-side heat-transfer quantities are listed, since the gas-side values have the same magnitude and opposite sign. The area error ε_A is reported as the absolute relative difference between $A_{\Gamma,\mathrm{eff}}$ and $A_{\Gamma,\mathrm{geo}}$. The table is used to assess both the agreement between the numerical results and the theoretical reference values and the resolution convergence of the interfacial heat-transfer statistic.

At coarse resolutions, both the end-time interfacial heat-transfer rate and the cumulative interfacial heat transfer still vary noticeably with refinement. However, once the particle spacing is reduced to $\Delta p \leq 0.004\,\mathrm{m}$, the cumulative transferred heat remains close to the theoretical value, with the difference generally within a few percent. The effective interfacial area also remains close to the geometric reference area. This indicates that the normal-projected pairwise flux measure provides a reasonable estimate of the macroscopic interface area for the initially flat interface. By contrast, the end-time interfacial heat-transfer rate is more sensitive to resolution and reaches a relatively stable range only when the particle spacing is refined to about $\Delta p = 0.0030$–$0.0028\,\mathrm{m}$.

This difference in convergence behavior is expected. The end-time interfacial heat-transfer rate depends directly on the local temperature gradient near the interface and is therefore more sensitive to the near-interface resolution. The

Table 5: Resolution sweep for the static tank using the free-surface-band interfacial statistic at the last output time ($t_{\mathrm{end}} \approx 3.40$–$3.42\,\mathrm{s}$, $\Delta t_{\mathrm{sample}} \approx 0.1\,\mathrm{s}$). The reference values are $\dot{Q}_{\Gamma,\mathrm{ref}}(3.4\,\mathrm{s}) = 2.31\,\mathrm{W}$, $E_{\Gamma,\mathrm{ref}}(0, 3.4\,\mathrm{s}) = 15.71\,\mathrm{J}$, and $A_{\Gamma,\mathrm{geo}} = \pi D^2/4 = 6.5144 \times 10^{-2}\,\mathrm{m}^2$.

Δp (m)	$N_L + N_G$	$\dot{Q}_\Gamma^{w,\mathrm{fs}}$ (W)	Err. $\dot{Q}$ (%)	$E_\Gamma^{w,\mathrm{fs}}$ (J)	Err. E (%)	$A_{\Gamma,\mathrm{eff}}$ (m^2)	ε_A (%)
0.0080	99661	3.33507	+44.38	11.3044	-28.04	0.061224	6.018
0.0060	238196	3.17213	+37.32	12.6369	-19.56	0.062354	4.283
0.0040	803991	3.37825	+46.24	15.2721	-2.79	0.062677	3.788
0.0036	1103670	3.04204	+31.69	15.1389	-3.64	0.063290	2.846
0.0034	1310070	2.98722	+29.32	15.3033	-2.59	0.063175	3.023
0.0032	1572575	2.88787	+25.02	15.2892	-2.68	0.063259	2.894
0.0030	1908596	2.40241	+4.00	16.1309	+2.68	0.063538	2.466
0.0028	2344293	2.34978	+1.72	15.5159	-1.24	0.063273	2.872
0.0026	2932950	2.39434	+3.65	16.1009	+2.49	0.063739	2.158
0.0024	3727459	2.34775	+1.63	15.5791	-0.83	0.063507	2.513
0.0022	4840196	2.36672	+2.46	15.8836	+1.10	0.063914	1.888
0.0020	6441165	2.34647	+1.58	15.4967	-1.36	0.063913	1.890
0.0016	12572266	2.33657	+1.15	15.5410	-1.08	0.063846	1.992

cumulative transferred heat, on the other hand, is obtained by time integration and is thus less sensitive to local fluctuations at any single output instant.

Considering the overall convergence behavior together with the total particle count and the computational cost, the resolution $\Delta p = 0.0028\,\mathrm{m}$ is adopted in the subsequent simulations as a practical compromise between thermal accuracy and computational efficiency.

The thermal diffusivities of the liquid and gas phases are

$$\alpha_L = \frac{k_L}{\rho_L c_{p,L}}, \qquad \alpha_G = \frac{k_G}{\rho_G c_{p,G}}.$$

Thus, at $t \approx 3.4\,\mathrm{s}$, the estimated thermal diffusion lengths,

$$\delta_m(t) \sim 2\sqrt{\alpha_m t},$$

are of the order of 1.3 mm on the liquid side and 7.7 mm on the gas side. This indicates that the liquid-side thermal boundary layer is already of millimeter order, which explains why quantitatively reliable prediction of interfacial heat transfer requires substantially finer resolution than the bulk hydrodynamic response.

4. Heat transfer in the Cryogenic Tank under different sloshing conditions

The equation of lateral excitation can be written as

$$X = A \cdot sin(2\pi f \cdot t), \tag{82}$$

$$a_x = -4\pi^2 f^2 A sin(2\pi f \cdot t). \tag{83}$$

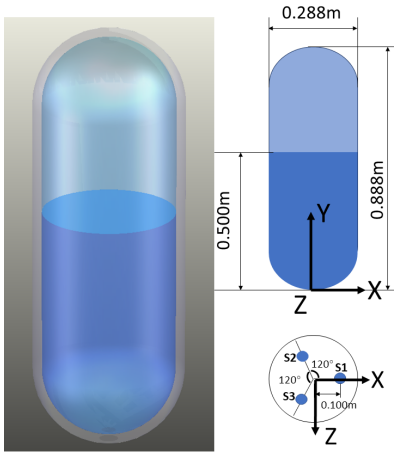


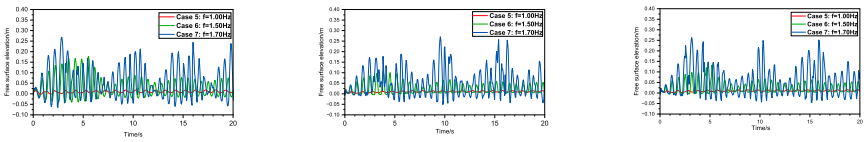
Figure 13: Cryogenic tank geometry and sensor locations

Properties	Values	
	Liquid methane	Nitrogen
Initial temperature T $[K]$	111.15	131.15
Thermal conductivity k $[W/(m \cdot K)]$	0.1846	0.01183
Specific heat capacity c_p $[J/(kg \cdot K)]$	3139.7	1040

Table 6: Thermal properties of liquid methane and nitrogen

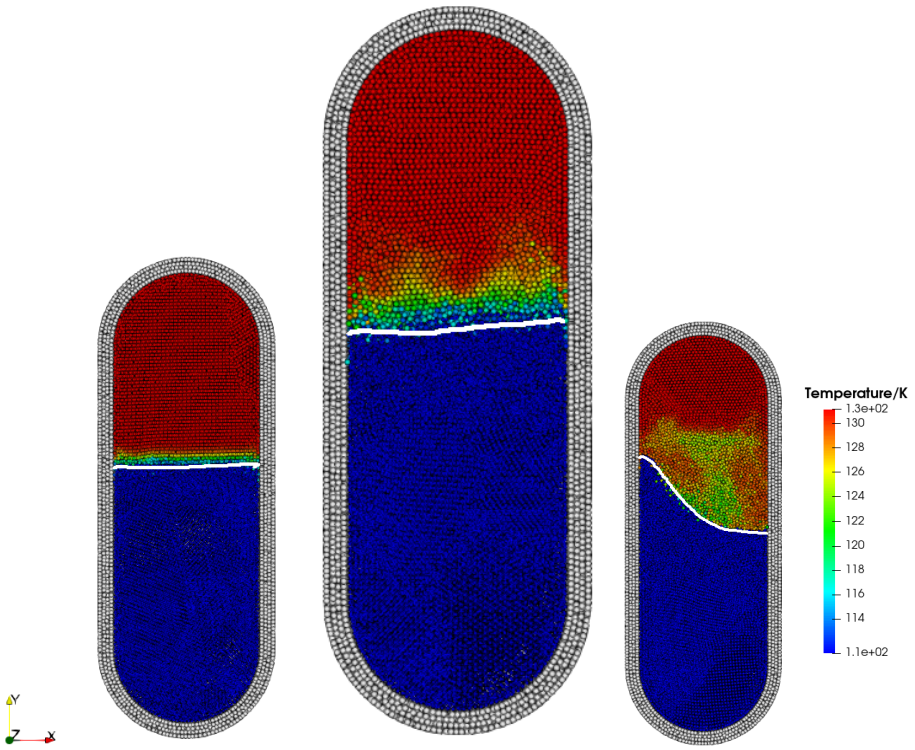
Case	Lateral excitation amplitude A [m]	Lateral excitation frequency f [Hz]
5	0.01	1.00
6	0.01	1.50
7	0.01	1.70

Table 7: Sloshing cases



(a) (b) (c)

Figure 14: Free-surface response for cryogenic tank cases 5–7



(a) (b) (c)

Figure 15: Temperature distribution in the tank at 20s

4.1. Identify non-resolved thin structures

4.2. level-set re-distance and reinitialize

5. The effect of baffles in inhibiting heat transfer

5.1. Identify non-resolved thin structures

5.2. level-set re-distance and reinitialize

6. Concluding remarks

In this paper, we have developed a level-set based self-clean pre-processing tool for particle-based methods. The present technique can automatically detect and clean-up the non-resolved tiny structures of 'dirty' geometries at certain grid resolutions. A number of numerical tests are demonstrated to validate the 'dirty' geometry cleaning-up ability of our pre-processing tool. The cleaning function of our method can not only identify and clean-up the non-resolved tiny structures but also automatically stop cleaning at the main body with the topological consistency. So as to improve the consistency of the topology structure while ensuring the integrity of the geometry to the greatest extent. Meanwhile, an efficient truly narrow-band technique of level-set data structure is exploited to enhance computational efficiency and reduce the memory usage. Numerical cases demonstrate that with the truly narrow-band data structure, the level-set operations can even achieve a more than tenfold speedup when high spatial resolution is applied. In addition, a level-set based 'static confinement' boundary condition is developed to complete the kernel support in the physics-driven relaxation process. By calculating the missing part of kernel support and storing to the background level-set cell center, the physics-driven relaxation process can not only obtain the full kernel support for particles close to the boundary but also accelerate the computation.

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CRediT authorship contribution statement

Yongchuan Yu: Investigation, Methodology, Visualization, Validation, Formal analysis, Writing - original draft, Writing - review & editing; **Yujie Zhu:** Investigation, Methodology, Formal analysis, Writing - review & editing; **Chi Zhang:** Investigation, Methodology, Formal analysis, Writing - review & editing; **Xiangyu Hu:** Investigation, Methodology, Supervision, Writing - review & editing; **Hao Ma:** Investigation, Writing - review & editing; **Oskar J. Haidn:** Investigation, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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